MATHLERIUM'S QUESTION · HAMID NADERI YEGANEH, "VIEW OF A BLACK HOLE FROM A PLANET"

How the black-hole equation is built

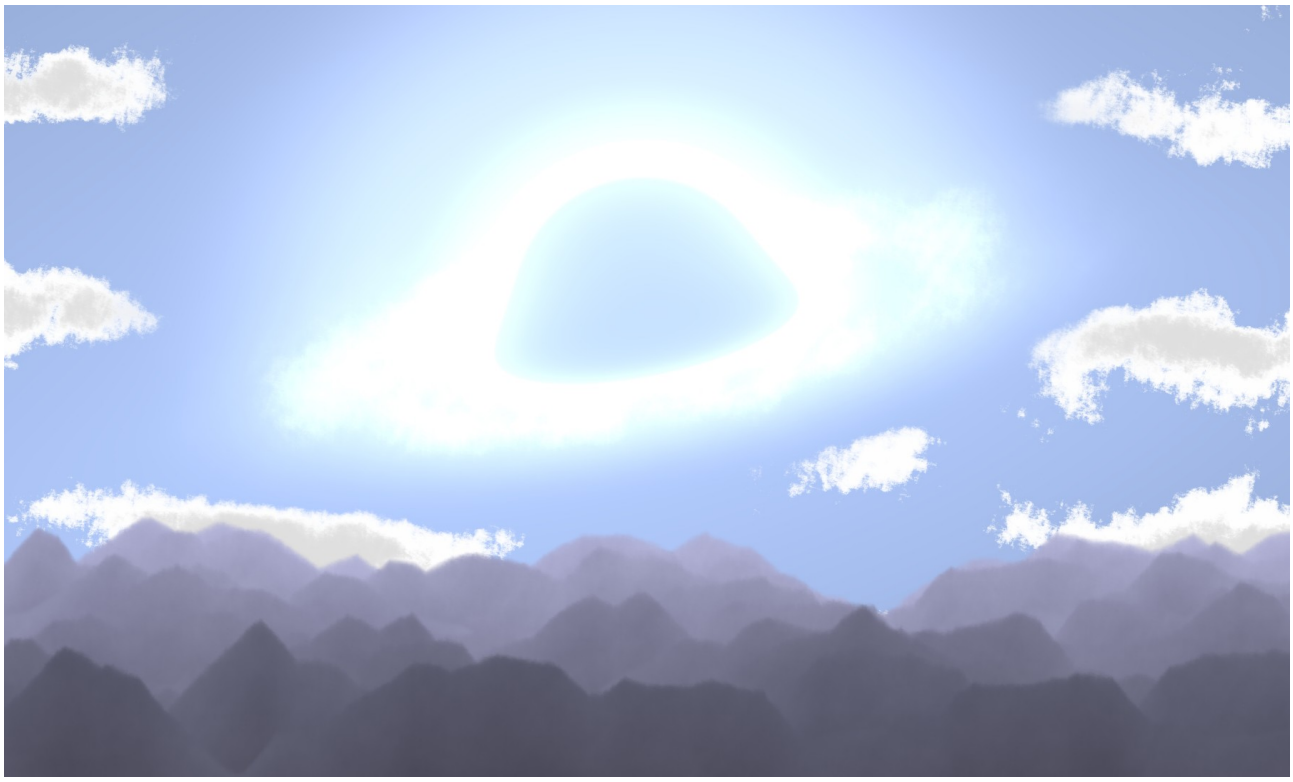
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AI-assisted: prepared with AI tools; every identity and render was checked by running it.

The formula is a picture written as algebra: a stack of a few simple operations. Shapes are level sets. Edges are soft steps. Glows are bands around a shape. Ridges are folds. Texture is a fractal sum. Depth is an occlusion chain. Colour is a quadratic in the channel number. Below are three demonstrations: the formula is rendered exactly, taken apart term by term, and then the same grammar builds a new picture from scratch.

1. The printed formula, rendered

Coded directly from the printed equations (pixel at row n, column m, with $x = (m - 1000)/500$ and $y = (501 - n)/500$), it reproduces the published picture. Nothing hidden: every element of the image is in the formula.



Our render of the full 2000×1200 image from the formula alone (24 s on a laptop).

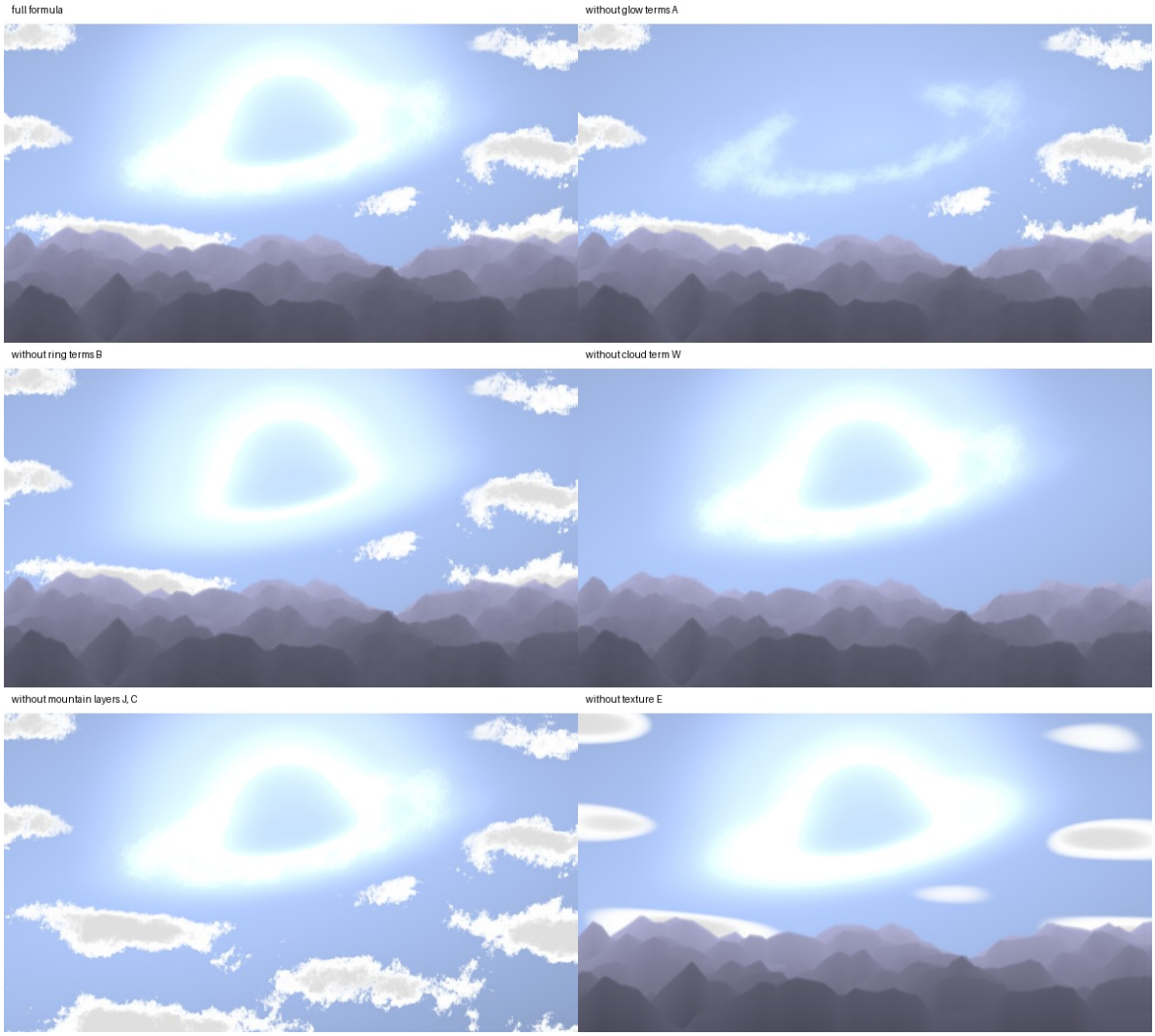
2. The grammar

Every term in the formula is one of these operations. The exact identities were checked numerically.

ROLE	OPERATION	IN THE FORMULA	CHECK
Edge	Soft step $S(z) = e^{-e^{-kz}}$: about 0 for $z < 0$ and about 1 for $z > 0$. Large k gives a crisp edge, small k a glow.	Almost every factor; k from 1000 to 3/4	matches a hard step to 4×10^{-5} away from the edge
AND	Adding inner exponentials multiplies steps: $e^{-e^a - e^b} = S(-a)\,S(-b)$	A_v, B_v, $J_{v,s}$	exact (10^{-16})
Shape	Level set of one function. $L(x, y)$ is a tilted ellipse whose lower half is weighted $10\times$ by a soft step across $y = x/5$; that one-sided weight makes it look lensed.	$L(x, y)$	removing the weight gives a plain ellipse
Glow	Band around a level set, $e^{-e^{v(\lvert L-0.3\rvert-0.05)}}$, at three sharpnesses (core, halo, haze)	A_{100}, A_5, $A_{3/4}$	ablation below
Ridge	Fold: $\arccos(\cos\theta)$ is a triangle wave; three folds at rates 4, 7, 17 with phase warps make a skyline	$J_{v,s}$	ablation below
Texture	Fractal sum: amplitudes $(9/10)^s$, frequencies $(5/4)^s$, angles from s^2. Roughness exponent $\ln(10/9)/\ln(5/4) = 0.47$.	$E(x, y)$, D_s	without E the clouds become smooth ovals
Depth	Painter's chain: layer s shows where no nearer layer does, $\prod_{u<s}(1 - J_u)\,J_s$; these weights and "nothing hit" add up to exactly 1	$C(x, y)$ and the $\prod$ factors	sums to 1 within 4×10^{-16}
Colour	Each coefficient is a quadratic in the channel index $v = 0, 1, 2$ fixed by three chosen values; $(v^2 + v + 14)/20$ gives R, G, B $= 0.70, 0.80, 1.00$	every v-polynomial	$(3v^2 - 3v + 28)/40 \to 0.70, 0.70, 0.85$
Output	$F(x) = \lfloor 255\, e^{-e^{-1000x}} \lvert x\rvert^{e^{-e^{1000(x-1)}}} \rfloor$ is $\lfloor 255 \cdot \mathrm{clamp}(x, 0, 1) \rfloor$ written with soft steps	F	differs at 14 of 400,001 points, all within 0.01 of 0 or 1

3. Taken apart

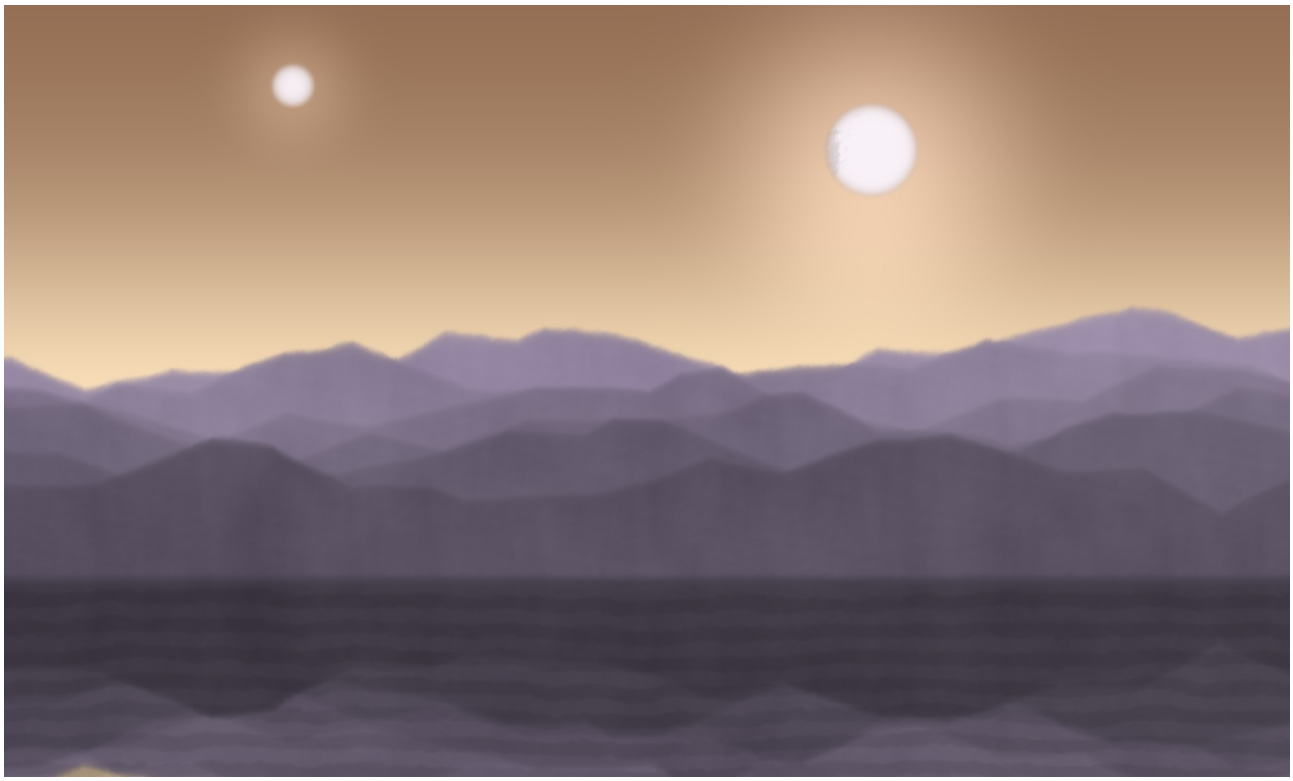
Switch off one group of terms and exactly one element of the picture disappears. Without the mountain layers you see the clouds that were behind them, so the depth order is real. Without the texture E, the clouds become smooth ovals: they are a blob pattern with fractal edges.



Full formula; then without the glow terms A, the ring terms B, the cloud term W, the mountain layers J and C, and the texture E.

4. The grammar, run forwards

If this is how the equations are made, the same operations should write a new picture to order. Here is one designed and written with nothing but the grammar above: two moons over a mountain lake at dusk. The lake is one more fold, $y \mapsto -\frac{39}{25} - y$, which reflects the whole scene.



New image, 2000×1200, generated entirely by the formula below. Re-implemented independently from this written formula, it matches in all 2,400,000 pixels.

For $m = 1, \ldots, 2000$ and $n = 1, \ldots, 1200$ the pixel colour is $\mathrm{rgb}\big(F(H_0), F(H_1), F(H_2)\big)$ at $x = (m - 1000)/500$, $y = (501 - n)/500$, with F as in the original and

$$H_v = (1-\lambda)\,P_v(x,y) + \tfrac{18}{25}\,\lambda\Big(1 - \tfrac{2}{25}e^{-e^{4\cos(70y+E/2)}}\Big)P_v(\tilde{x},\tilde{y}), \qquad \lambda = e^{-e^{-80\left(-\frac{39}{50}-y\right)}},$$

$$\tilde{x} = x + \tfrac{3}{500}\cos\big(50y + \tfrac{E}{2}\big), \qquad \tilde{y} = -\tfrac{39}{25} - y,$$

$$P_v = \sum_{s=1}^{6}\Big(\prod_{u=1}^{s-1}(1-J_u)\Big)J_s\Big(\tfrac{3v^2-5v+22}{40}\big(\tfrac{11}{20} + \tfrac{9s}{100}\big) + \tfrac{E}{120}\Big) + \Big(\prod_{s=1}^{6}(1-J_s)\Big)S_v,$$

$$J_s = e^{-e^{-120(r_s-y)}}, \quad r_s = -\tfrac{33}{50} + \tfrac{3s}{40} + \tfrac{\arccos\cos(3x + \frac{7}{10}\cos(2x+s^2)) + s^2)}{5\pi} + \tfrac{7\arccos\cos(9x+\cos(5x+3s^2)+2s^2)}{100\pi} + \tfrac{E}{250},$$

$$S_v = \Big((1-t)\tfrac{-v^2-3v+40}{40} + t\,\tfrac{v^2-7v+22}{40} + G(1-\mu_1-\mu_2)\Big)(1-\mu_1-\mu_2) + \tfrac{v^2-2v+39}{40}\Big(1 - \tfrac{3}{25}e^{-e^{6(E/2+2/5)}}\Big)(\mu_1+\mu_2),$$

$$t = e^{-e^{-3(y-\frac{3}{20})}}, \quad \mu_1 = e^{-e^{400(M_1-\frac{1}{50})}}, \quad \mu_2 = e^{-e^{900(M_2-\frac{1}{250})}}, \quad G = \tfrac{7}{20}e^{-e^{4(\sqrt{M_1}-\frac{7}{20})}} + \tfrac{1}{5}e^{-e^{8(\sqrt{M_2}-\frac{3}{25})}},$$

$$M_1 = \big(x - \tfrac{7}{10}\big)^2 + \big(y - \tfrac{11}{20}\big)^2, \qquad M_2 = \big(x + \tfrac{11}{10}\big)^2 + \big(y - \tfrac{3}{4}\big)^2,$$

$$E(x,y) = \sum_{s=1}^{40}\Big(\tfrac{9}{10}\Big)^s \cos\Big(\big(\tfrac{5}{4}\big)^s\big(\cos(3s^2)x + \sin(3s^2)y\big) + 3\cos\Big(\big(\tfrac{5}{4}\big)^s\big(\cos(5s^2)x + \sin(5s^2)y\big)\Big) + \cos 4s\Big).$$

In $P_v(\tilde{x},\tilde{y})$ the texture E is also evaluated at the reflected point.

What this shows

Every term of the black-hole formula is one of eight simple operations, and each group of terms draws exactly one element of the picture. Run forwards, the same operations write a new formula for a new scene that renders exactly as designed. That is a working, checkable account of how equations like these are made: design the scene as shapes, edges, glows, folds, texture and layers; write each as its operation; then tune the constants by eye. His constants are round fractions such as 21/20, 3/10 and 9/10, which looks hand-tuned rather than fitted.

What it cannot show is his private process: the order he works in, or how many trials each constant took. Only he can say that.

Scripts: `render.py` (the original), `anatomy.py` (identities and ablations), `new_scene_v2.py` (the new image), `transcription_check.py` (re-implements the written formula and compares all 2,400,000 pixels: 0 differences).